Enrico **Magazzino**

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PROFILE

PhD researcher at the University of Padova specialising in Earth observation-driven monitoring of regenerative agriculture, with a focus on integrating Sentinel-1 SAR, Sentinel-2 multispectral data, and UAV-based LiDAR and photogrammetry within machine learning frameworks. Research interests span precision agriculture, landscape biodiversity indicators, and the ecological dimensions of land-use change, grounded in a background in natural science, environmental biology and grassland ecology. Experienced in the full research pipeline: from UAV field campaigns and satellite data processing to workflow development in Python, scientific writing, and international conference dissemination.

RESEARCH EXPERIENCE

September 2022 – July 2023

Researcher

Utrecht University, Utrecht, The Netherlands

Study of the formation of regular vegetation patterns in drylands:

- Drought experiments with *Paspalum vaginatum* L. under laboratory-controlled conditions.
- Remote sensing imaging analysis (RGB, Multispectral, and Thermal imaging).
- Mathematical modelling, data analysis, and scientific writing (papers, protocols, reports).
- Invited talk at the NWO Life 2023 conference.

EDUCATION

November 2024 – Ongoing

PhD in Agritech Smart Mapping (Industrial Doctorate)

University of Padova, 40th Cycle · Scholarship funded by PNRR MD 630/2024

Thesis theme: *Earth Observation–Driven Monitoring of Regenerative Agriculture: Satellite–UAV Fusion for Practice Verification and Landscape Biodiversity Indicators*

- Advanced research on Sentinel-1 SAR and Sentinel-2 multispectral data integration with machine learning for monitoring regenerative agricultural practices; UAV campaigns using LiDAR and multispectral photogrammetry to extract biodiversity indicators.
- Management of GeoServer and WebGIS platforms (MapStore) for the RurAll project.
- UAV field operations for high-resolution remote sensing surveys.
- Workflow development in Python, Google Earth Engine, QGIS, and ArcGIS for large-scale geospatial analysis.
- Supervision of a Master's thesis.
- Teaching assistant: Remote Sensing and GIS Basics courses (Bachelor's and Master's level).
- Period at company (**Rurall S.p.a.**):
  - Automated pipeline for Sentinel-1 and Sentinel-2 indices ingestion and GeoServer/MapStore visualisation.
  - LiDAR surveys on 10 farmlands for biodiversity assessment (client: NaturaSì).
- Period abroad: *CIDE – Desertification Research Centre, Spanish National Research Council (CSIC)*:
  - Mapped additional areas for biodiversity with LiDAR; developed automated extraction software for structural vegetation elements, blue-green infrastructure, functional/structural proxies, and microtopography/hydrology.

October 2023 – September 2024

Second-Level Master's Degree in GIScience and Remote Piloted Systems for the Integrated Management of Territory and Natural Resources

University of Padova · EQF Level 8

Field of study: Geo-information and New Technologies for Sustainable Agriculture

- Skills acquired: GIScience, cartography, mobile-GIS, reference systems, GIS software and geographic data formats, territorial interpretation, geodatabases and associated programming languages, satellite and drone remote sensing, WebGIS, remote piloted systems, participatory GIS, and geoinformatics.
- **Thesis:** *Interplay between LST spatial variability and vegeto-productive anomalies. Case studies on olive trees and vineyard crops in the Parco Regionale dei Colli Euganei area.*

September 2018 – December 2021

Master's Degree in Biosciences (Environmental Biology; Ecology and Natural Resource Management)

Utrecht University · EQF Level 7

- MSc internship – Research profile (9 months): *"The effect of drought on plant soil microbiomes: evidencing potential novel benefits and application in land management."* (Vicenza, Italy; March–November 2021)
- MSc internship – Major project (9 months): *"Effects of drought on plant traits, diversity, community composition and ecosystem functioning in a semi-natural grassland in the Netherlands."* (Utrecht, January–October 2019)
- Short project (1 month): short-time recovery after leaf harvesting of *Geonoma baculifera* (Poit.) Kunth. (Utrecht, February–March 2020)
- Writing assignment (5 weeks): *"Birds abundance surveys in forest habitats: comparing traditional methods with acoustic recording — a meta-analysis."* (Utrecht, April–May 2020)
- Major courses: Management of Natural Resources in Context; Ecology of Natural Resources; Tropical Ecology (Vrije Universiteit Amsterdam) with field trip to Tiputini Biodiversity Station, Ecuador (December 2019).

October 2012 – December 2017

Bachelor's Degree in Natural Science (Environmental Sciences, L-32)

University of Padova · EQF Level 6

- Major courses: Zoology, General Botany, Physical Geography and Geomorphology, Systematic Botany and Vegetation Ecology, Environmental Assessment of Territory, Environmental Impact, Ecological Management and Restoration, Plant Physiology, Ethology, General Physiology.
- Erasmus Plus exchange, University of Alicante.
- Bachelor's thesis research carried out within the NASSTEC Marie Curie Multi-partner ITN European Project at MUSE (Science Museum of Trento):
  - "Preliminary germination and post-germination studies of 6230 Species-rich Nardus grasslands (Natura 2000) species currently not commercially available for restoration in the Trentino region."

PUBLICATIONS

- Native Seed Ecology, Production & Policy – Advancing knowledge and technology in Europe. (*Royal Botanic Gardens, Kew*)
- Using Landsat LST Data to Predict Vineyard Productivity Anomalies: A Case Study in the Euganean Hills wine region, Italy. *Awarded 2nd place, SIFET Conference, Brindisi (2025); forthcoming on SIFET website.*
- Davin, A., von Hardenberg, J., Berghuis, P. M., Mayor, Á. G., Magazzino, E., Rietkerk, M., ... & Baudena, M. (2026). Guerrilla clonal growth strategy leads to amorphous pattern formation in a drylands vegetation model. *Ecological Modelling*, *515*, 111510. DOI: 10.1016/j.ecolmodel.2026.111510
- Opposing effects of rainfall and nutrient shifts on temperate grassland stability via asynchrony, resistance, and diversity. *Under review — Journal of Ecology.*

SKILLS

Programming & Data: Python (professional), R (professional), Google Earth Engine, MATLAB (basic), LaTeX.

Geospatial: QGIS, ArcGIS, GeoServer, MapStore, WebGIS, GPS/RTK/GCP workflows.

Remote Sensing: Sentinel-1 SAR (sarmap MAPscape and SNAP), Sentinel-2 (Copernicus Data Space), UAV LiDAR, multispectral and RGB photogrammetry (Agisoft Metashape, DJI Terra and others).

Other: Microsoft Office, Adobe Creative Suite, drone piloting licence (SAPR), scientific writing, conference presentations.

LANGUAGES

Italian (native) · **English** (C2 – professional) · **Spanish** (B2 – professional working) · **French** (A2 – basic)

ACTIVITIES & INTERESTS

Literature · Environmental conservation · Art · Climbing · Hiking · Skiing

Updated: June 2026

Reference contacts available upon request.

Enrico Magazzino